

Qianqian Li

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Education

Northeastern University, Shenyang, China[🔗]

Sept 2023 – Jun 2027

Bachelor of Artificial Intelligence, School of Computer Science and Engineering

- **GPA:** 88.4/100 (Overall), 92/100 (Major/Junior)
- **Distinctions:** Project 985, Project 211, Double First-Class University

Nanyang Technological University (NTU), Singapore[🔗]

Aug 2024

Visiting Student, Artificial Intelligence Summer Program

Fully-funded visiting program focused on Artificial Intelligence. Engaged in intensive daily coursework and delivered a comprehensive final project presentation.

Key Courses: Machine Learning (97), Computer Graphics (94), Discrete Math (92), Computer Systems (93), Compiler Construction (91), Numerical Analysis (91), Linear Algebra (93), Image Processing (95)

Research Interests

My research centers on **3D Spatial Understanding, Environmental Perception, and Cognitive AI**. I aim to bridge physical and digital worlds through intelligent AR environments and develop **Cognitive Digital Twins** to preserve expert human reasoning—enabling AI to push the boundaries of complex scientific discovery.

Publications

A time–frequency graph fusion framework for Major Depressive Disorder diagnosis in multi-site rsfMRI data[🔗]

Journal of Affective Disorders, 2026

Co-author

Research Experience

University of North Carolina at Chapel Hill – Research Intern

Aug 2025 – Present, Remote

Advisor: Prof. Mingyu Ding[🔗]

- Led a scalable simulation pipeline for generating 3D articulated objects with precise physics for robotic tasks.
- Parsed MuJoCo XMLs for rigid body assembly and generated 3D assets via Rodin to build robust spatial environments.
- Migrated the physics framework from iOS/Swift to Unity, resolving complex multi-collider bugs.

NEUDM, Northeastern University, China[🔗] – Research Assistant

Oct 2024 – Present

Advisor: Prof. Xiaocui Yang[🔗]

- Enhanced reasoning capabilities of small multimodal models for sentiment analysis on Twitter 2015 and 2017 datasets.
- Implemented knowledge distillation from LLMs and random-masking reasoning strategies to optimize model architecture.

Industry Experience

Lion Rock AI Lab, China Merchants Group – Robotics Intern

Jan 2026 – Mar 2026

Mentor: Zhelin Lan

- Configured environments (Ubuntu, Docker, ROS) for robot-server communication, and deployed SLAM algorithms for remote mapping.
- Researched state-of-the-art literature on human-following in occluded environments and participated in field deployments.
- Built an Automatic Speech Recognition (ASR) testing pipeline for museum robots using LLM APIs to process historical texts.

MOFA (Shenyang) Info Tech Co., Ltd. (X-Friend) – Technical Member

Sept 2024 – Present

Mentors: Prof. Zhiliang Zhu, Hongyan Lv

- Developed a backend system for photo-to-avatar conversion, optimizing DALL · E 3 prompts for high identity consistency.
- Employed prompt constraints to resolve character distortion artifacts commonly seen in Stable Diffusion baselines.
- Earned the Excellence Award at the Global Semifinals of the Zhongguancun Talents Maker Contest.

Extracurricular Activities

Student Association for Science and Technology – Dept. Head

Sept 2024 – Jul 2025

- Organized two major campus-wide events (500+ participants each) to foster student engagement.

Awards

- **National "Good" Project Rating** (Core Technical Member) Mar 2025
- **Yao Tianshun Scholarship** (Top 0.5%, 3/600 students) Oct 2024
- **Second-Class University Scholarship** (Top 10%) Sept 2024

Projects

Intelligent AR Companion: Interactive 3D Avatar in Real-world Environments

Nov 2025 – Mar 2026

- **AR & Animation:** Animated a 3D Unity avatar (lip-sync, gaze) and embedded it into an iOS app, utilizing ARKit for spatial occlusion.
- **RAG & Voice:** Built a microservice via Chroma DB for lore retrieval. Deployed DeepSeek API for dialogue and CosyVoice 2.0 for timbre-cloned voice synthesis.
- **Virtual Vision:** Implemented a virtual vision system using 3D Gaussian Splatting from mobile images, perfectly aligning 3D scenes with real-world coordinates.

Autonomous Multimodal Robot Car

Jan 2025 – Jun 2025

- Designed an autonomous multimodal robot capable of processing visual/auditory inputs and communicating asynchronously with a server.
- Implemented an AI decision loop via multimodal models to analyze live visual feeds and autonomously determine navigation parameters.
- Integrated text-to-speech for situational responses and minimized hardware bottlenecks via asynchronous task streaming.

Huffman Compression Web Service

Dec 2024

- Engineered a full-stack compression platform with a custom C++ server (macOS native libraries) and an HTML/JS frontend.
- Implemented Huffman coding for efficient, lossless compression/decompression of text, files, and images (up to 10MB).
- Resolved character encoding and binary stream bugs. Delivered a highly praised defense, resulting in an invitation for system research.

Architectural 3D Modeling & Printing

Mar 2024 – May 2024

- Reconstructed a print-ready 3D architectural model (interior/exterior) of an anime residence in Blender, leveraging hand-drawn sketches.
- Operated 3D printing hardware to fabricate the physical model, iteratively testing wall thickness and custom structural supports.

Technologies

Languages: Python, C/C++, Swift, HTML/CSS, JavaScript, SQL, MATLAB, LaTeX

Technologies & Frameworks: PyTorch, Unity, ROS, MuJoCo, SLAM, Docker, ARKit, RAG, Chroma, OpenCV, Git